

1.6

$$27 - 27 - 27 + 5 = -22$$

1.10

$$x = t^3 - 3t^2 - 9t + 5$$

$$v = 3t^2 - 6t - 9$$

$$= 3(t-3)(t+1)$$

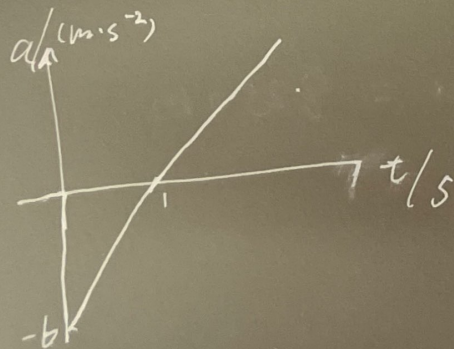
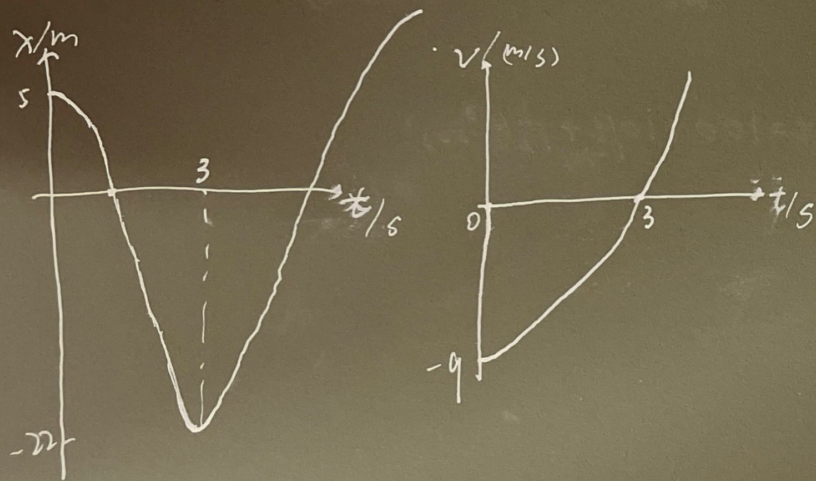
$t > 3$  s 时, 向  $x$  轴正向运动

$0 < t < 3$  s 时, 向  $x$  轴负向运动

$$a = 6t - 6 = 6(t-1)$$

$1 < t < 3$  s 时, 减速

$0 < t < 1$  s 或  $t > 3$  s 时, 加速



$$a = -2x, \quad x=0 \text{ 时}, v_0 = 4$$

$$\frac{d^2x}{dt^2} + 2x = 0, \quad r^2 = -2, \\ r = \pm \sqrt{2}i$$

$$\therefore x = C_1 \cos(\sqrt{2}t) + C_2 \sin(\sqrt{2}t)$$

$$t=0 \text{ 时}, x = x_0 = C_1$$

$$a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$$

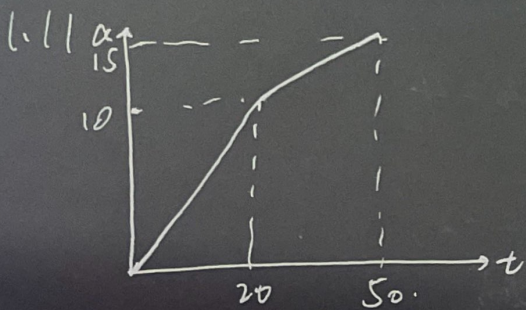
$$\therefore v \frac{dv}{dx} = -2x$$

$$\Rightarrow \frac{1}{2}v^2 = -x^2 + C$$

$$x=0, v_0=4 \Rightarrow C=8$$

$$\therefore v^2 = -2x^2 + 16$$

$$v = \sqrt{16 - 2x^2}$$



0 ~ 20 s 时,  $v = \int a dt = \int \frac{1}{2} t dt = \frac{t^2}{4} + C$ .

~~$a = 0.5 \text{ m/s}^2$~~

$a = 0.5t \quad t=0, v_0=0 \quad \therefore C=0$ .

$\therefore v = \frac{1}{4} t^2 \quad (0 < t < 20 \text{ s})$ .

20 s ~ 50 s 内,  $a = 10 + \frac{5}{30} t = 10 + \frac{1}{6} t \quad (\text{m/s}^2)$ .

~~$v = \frac{1}{4} \times 20^2 + \int_{20}^t a dt = 100 + 10(t-20) + \frac{1}{12}(t-20)^2$~~   $v(t) = \frac{1}{4} \times 20^2 + \int a dt = 100 + 10(t-20) + \frac{1}{12}(t-20)^2$

~~$t=50 \text{ s 时 } v = 100 + 500 + \frac{1}{12} \times 2500 =$~~

$v_{\text{末}} = \frac{1}{4} \times 20^2 + \frac{50-20}{2} \times (10+15) = 475 \text{ m/s}$ .

$0 < t < 20 \text{ s}, h_1 = \int_0^{20} v dt = \frac{1}{12} t^3 \Big|_0^{20} = \frac{1}{12} \times 8000 \text{ m} = \frac{2000}{3} \text{ m}$

$\therefore h_2 = \int_{20}^{50} (100 + 10(t-20) + \frac{1}{12}(t-20)^2) dt = \int_0^{30} (100 + 10u + \frac{1}{12} u^2) du = 8250 \text{ m}$

$\therefore H = h_1 + h_2 = 8916.67 \text{ m}$

1.24 (1) ~~在柱面上运动的自然法~~ <sup>直角坐标</sup>  $x_m = L_{Am} \cos \theta, y_m = L_{Am} \sin \theta$ ,  
 $L_{Am} = 2R \cos \theta. \quad \therefore \begin{cases} x_m = 2R \cos^2 \theta, -R = R \cos 2\theta = R \cos(2\omega t) \\ y_m = R \sin 2\theta = R \sin(2\omega t) \end{cases}$   
 ~~$x_m^2 + y_m^2 = R^2 \Rightarrow 4R^2 (\cos^4 \theta + \sin^2 \theta \cos^2 \theta) = R^2 \Rightarrow R \sin(2\omega t)$~~   
 自然法:  $v = 2\omega R$  切向.  $a = a_n = \frac{v^2}{R} = 4\omega^2 R$  指向 O

(2) 求  $A_m$  即可:  $A_m = 2R \cos \theta = 2R \cos(\omega t) = r$

$v_{\text{相}} = \frac{dr}{dt} = -2\omega R \sin(\omega t)$

$a_{\text{相}} = \frac{dv_{\text{相}}}{dt} = -2\omega^2 R \cos(\omega t)$